



Hosiery Knitting

👤 Created by	© Gustavo Vieira
🕒 Created time	April 23, 2026 12:02 AM
☰ Category	Methodology doc
👤 Last edited by	© Gustavo Vieira
🕒 Last updated time	April 23, 2026 12:07 AM
👥 Reviewers	 Laurent  Vincent  Kishore Ganesan

1. Introduction

1.1 Purpose of the study

This document describes the life cycle inventory (LCI) model for hosiery knitting as part of the Carbonfact Textile LCA Database. Hosiery knitting refers to the production of socks, stockings, and similar small-diameter tubular knitted goods on single-cylinder or double-cylinder circular machines. The model follows the same methodological principles applied across all knitting datasets in this database, ensuring coherence in the treatment of electricity, capital goods, indirect energy, and material losses.

1.2 System boundary

The system boundary covers gate-to-gate hosiery knitting: from yarn at the creel to finished hosiery fabric at the mill gate. **Included:** direct electricity, indirect energy, lubricant and waste mineral oil (inline auxiliary), capital goods and EoL (inline auxiliary), and building infrastructure. **Excluded:** fibre and yarn production, textile and material waste (handled externally via the IOR — no inline yarn-loss or textile-waste exchange in the dataset), sewing, dyeing, finishing, and packaging.

Functional unit: **1 kg of hosiery fabric** (average of mixed cotton and synthetic yarns). Background: ecoinvent 3.12, system model: cut-off by classification.

Note on sources: machine performance data are drawn from manufacturer catalogues (Lonati, Santoni, Sangiacomo, Busi Giovanni) and academic sources. Some supporting references (K&G Garment, Accio) are commercial or grey literature and are used only for indicative benchmarking.

2. Process Description

Hosiery machines are small-diameter single-cylinder or double-cylinder circular knitting machines designed for automatic production of socks, stockings, and tights. The **Lonati GE/GL series** and **Busi Giovanni 500 series** are the principal references for this model, covering the 230–320 kg machine weight range and a productivity of 300–800 pairs per day (~0.8–1.0 kg fabric per hour for standard socks).

Modern hosiery machines with automatic toe closure achieve defect rates below 2% and near-zero cutting waste. Yarn losses in hosiery arise from thread breaks, machine setup, and the small amount of separation thread used to distinguish consecutive articles on the machine. Total material loss is typically 2–5% (Laursen et al., 2007; Accio, 2023).

Specific electricity consumption is **0.5–1.2 kWh/kg** for machine electricity alone, derived from Lonati GK series data (~0.5 kW running load at 0.8–1.0 kg/h → 0.5–0.65 kWh/kg, consistent with Maity et al., 2021). The adopted default is **1.2 kWh/kg** — the upper end of this range, applied conservatively.

3. Life Cycle Inventory

The inventory below represents the average scenario. Yarn losses and waste-yarn treatment are excluded and handled by the caller via the IOR (Section 4). Auxiliary waste flows — lubricant EoL and machine EoL — are modelled inline.

Flow	Amount	Unit	Dataset	
Technosphere inputs				

Flow	Amount	Unit	Dataset	
Lubricant, average	0.008	kg	`lubricating oil production {RoW} \	cut-off` — Sandin et al. (2019), p. 45: upper bound; consistent with 0.05–0.10 L/kg from Lonati technical brochures (2023)
Waste mineral oil (lubricant EoL)	0.008	kg	`market for waste mineral oil {RoW} \	cut-off`
Electricity				
Machine electricity (direct)	1.2	kWh	`market group for electricity, medium voltage {GLO} \	cut-off` — upper bound of 0.5–1.2 kWh/kg range (Lonati GK series; Maity et al., 2021, p. 235)
Indirect energy (HVAC, compressors, lighting, auxiliaries)	4.46	kWh	`market group for electricity, medium voltage {GLO} \	cut-off` — shared default: $2.4 \div 0.35 \times 0.65$
Capital goods				
Building machine (capital good)	1.33×10^{-6}	unit	`market for building machine {GLO} \	cut-off` — $300 \text{ kg} \div 720,000 \text{ kg} \div 300 \text{ kg/unit}$
Building infrastructure	1.61×10^{-5}	m ²	`market for building, hall {GLO} \	cut-off`
Waste flows (auxiliary only)				
EoL machine — steel (85%)	3.54×10^{-4}	kg	`market for waste steel	cut-off` — $255 \text{ kg} \div 720,000 \text{ kg}$

Flow	Amount	Unit	Dataset	
			{RoW} \	
EoL machine — other (15%)	6.25×10^{-5}	kg	`treatment of waste electric and electronic equipment, shredding {GLO} \	cut-off`

Note: the foreground dataset does not include yarn input or textile-waste flows. Yarn losses (2% default) are handled externally via the IOR by the caller.

4. Capital Goods, Building Infrastructure and Indirect Energy

Shared methodology: Capital Goods & End-of-Life (Machinery) in LCA — Full modelling framework and default values. This section states only hosiery-specific inputs.

Shared methodology: Building Infrastructure in LCA — Full allocation framework. Section 4.2 states only the hosiery-specific value.

4.1 Machine parameters

The reference machine is the **Lonati GE616DF / GE516DF family** (300 kg), representing standard single-cylinder hosiery knitting machines (Lonati/Santoni technical catalogues, 2023). Productivity of 10 kg/h is derived from 300–400 pairs per day at an average sock mass of 50–70 g.

Parameter	Value	Source / note
Machine model	Lonati GE616DF/GE516DF	Lonati/Santoni brochures (2023)
Machine mass (kg)	300	Lonati/Santoni (2023)
Service life (years)	15	Average scenario
Operating hours per day	16	Average scenario

Parameter	Value	Source / note
Operating days per year	300	Average scenario
Productivity (kg/h)	10	Derived: 300–400 pairs/day at ~50–70 g/pair
Lifetime output Q (kg)	720,000	$15 \times 16 \times 300 \times 10$
Machine allocation (kg/kg fabric)	4.17×10^{-4}	$300 \div 720,000$
ecoinvent units	1.33×10^{-6}	$300 \div 720,000 \div 300 \text{ kg/unit}$

Machine EoL: 85% steel (255 kg) → **$3.54 \times 10^{-4} \text{ kg/kg fabric}$** → market for waste steel {RoW} | cut-off ; 15% other (45 kg) → **$6.25 \times 10^{-5} \text{ kg/kg fabric}$** → treatment of waste electric and electronic equipment, shredding {GL0} | cut-off .

4.2 Building infrastructure

Building infrastructure is allocated using the Lonati GE616DF3 envelope (1,273 mm × 1,148 mm ≈ 1.46 m²) with a higher cell factor than other knitting technologies to account for multi-machine cell layout, shared creel systems, and common aisles. This yields a building intensity consistent with the seamless knitting default: **$1.61 \times 10^{-5} \text{ m}^2/\text{kg}$** is adopted for hosiery, given comparable normalised throughput per allocated space.

ecoinvent activity: market for building, hall {GL0} | cut-off (unit: m²).

4.3 Indirect energy

Shared methodology: Indirect Energy in LCA — Shared Methodology (Knitting)
— Full derivation, evidence base and subsystem distribution.

Indirect energy is modelled at **4.46 kWh/kg**, consistent with all knitting processes in this database (ADEME, 2024: $2.4 \div 0.35 \times 0.65 = 4.46 \text{ kWh/kg}$). For hosiery, relevant auxiliary systems include compressed-air lubrication delivery, suction for yarn tensioning, and illumination of multi-machine cells.

5. Input–Output Ratio (IOR)

Definition: $IOR = 1 / (1 - L)$, where L is the yarn loss rate (as a fraction). For every 1 kg of finished hosiery fabric, the upstream yarn input must be scaled by IOR.

How the loss rate is derived

Modern hosiery machines with automatic toe closure achieve defect rates below 2%. The **2% default** is consistent with Accio (2023) industry benchmarks and the Laursen et al. (2007) range for circular knitting. Yarn losses include thread breaks, machine setup waste, and separation thread consumption.

IOR value

Scenario	Loss (%)	IOR (kg/kg)	Input required (kg)	Loss per kg output (kg)
Default	2%	1.0204	1.0204	0.0204

How to apply: for 1 kg of hosiery fabric, multiply upstream yarn input by **1.0204**. The 0.0204 kg of lost yarn is routed by the caller to `market for waste yarn and waste textile {RoW} | cut-off`. No inline yarn-loss or textile-waste exchange is included in the dataset itself.

6. Inferences and Limitations

The direct electricity value of 1.2 kWh/kg represents the upper bound of the empirical range. The lower end (0.5–0.7 kWh/kg from Lonati machine data at operating load) suggests that the adopted value is conservative and may overestimate machine electricity for efficient modern hosiery lines. Primary sub-metered measurements from operating hosiery facilities would allow replacement with a more precise value.

The indirect energy estimate (4.46 kWh/kg) dominates the inventory and carries ±30–50% uncertainty, as for all knitting processes in this database. Hosiery facilities typically run many machines simultaneously in a shared space, which may affect the per-machine auxiliary load differently from single-machine cells assumed in the derivation.

Capital goods contribute less than 1% of process GWP and are not a material source of uncertainty.

7. References

- Accio. (2023). *Socks knitting machine benchmark*.
https://www.accio.com/plp/socks_knitting_machine (commercial web source; used for indicative productivity benchmarking only)
- ADEME & Ministère de la Transition Écologique. (2024). *Ecobalyse methodological guide — Textiles and footwear* (Version 1.0). Paris, France.
- Fazio, S., Castellani, V., & Sala, S. (2020). *Guide for EF-compliant datasets*. European Commission, JRC.
- GIZ. (2023). *Sustainable apparel report — Cutting waste reduction*. Deutsche Gesellschaft für Internationale Zusammenarbeit.
- Hasanbeigi, A., & Price, L. (2015). *Journal of Cleaner Production*, 95, 30–44.
<https://doi.org/10.1016/j.jclepro.2015.02.079>
- K&G Garment. (2025). *What is seamless knitting?*
<https://kggarment.com/blogs/news/what-is-seamless-knitting-a-complete-guide-for-brands>
- Laursen, S. E. et al. (2007). *EDIPTEx — Environmental assessment of textiles*. Danish Environmental Protection Agency.
- Lonati S.p.A. (2023). *GE516DF / GE616DF technical catalogue*. Lonati/Santoni.
- Maity, S., Singha, K., & Singha, M. (2021). In *Advanced Knitting Technology* (pp. 225–260). Woodhead. <https://doi.org/10.1016/B978-0-12-821290-3.00009-0>
- Sandin, G., Roos, S., & Johansson, M. (2019). *Environmental impact of textile fibres*. Mistra Future Fashion, IVL.
- Van der Velden, N. M., Patel, M. K., & Vogtländer, J. G. (2014). *Int. J. LCA*, 19(2), 331–356. <https://doi.org/10.1007/s11367-013-0626-9>
- Weidema, B. P. et al. (2013). *Data quality guideline for ecoinvent v3*.
<https://doi.org/10.25678/0006HH>